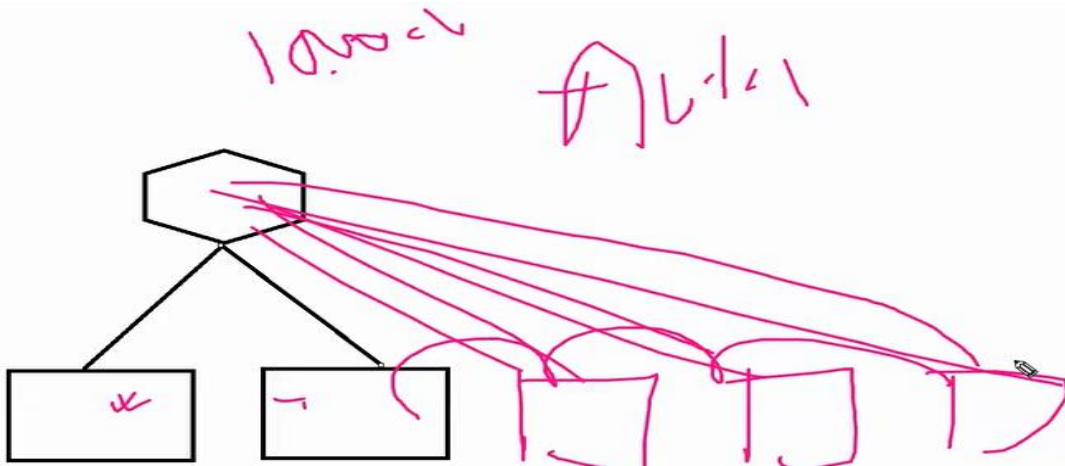


## Virtual Machine Scale Set (VMSS)

✓ **Horizontal Scaling** (Web, App etc)

✓ We are adding up machines here

horizontal scaling : adding machine automatically



✓ **Vertical Scaling** (DB & File servers)

✓ We are upgrading machines here

Vertical scaling

upgrade your size

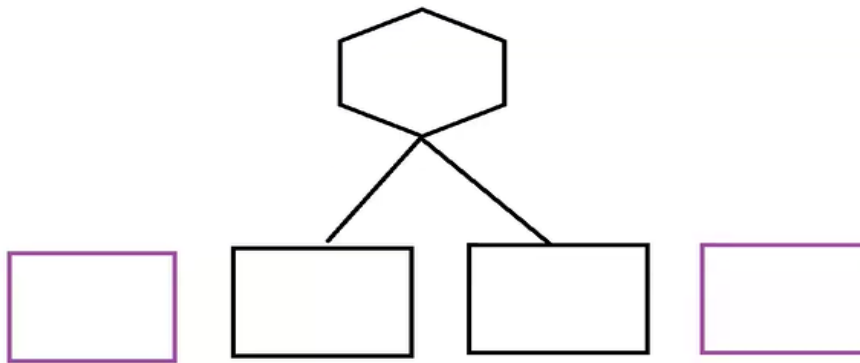
standard b1\_S : 1CPU and 1GB RAM

2cpu and 8 GB RAM

✚ Why we don't use vertical scaling on applications & Web bcz it has frontend pages and no data, when the traffic increases machines automatically created by (Horizontal) and upgraded(vertical), when traffic down machines will be deleted then upgraded data also deleted.

✚ Here we lose data that's why we don't use vertical scaling on applications and Web.

✓ In Horizontal scaling we use Custom images only, because it should be identical to existing machines.



## ❖ VMSS

- Vertical Scaling

- Upgrading size

✓ Created VM with 1cpu,8GB

✓ Let's Upgrade this to 1cpu,16GB.



Size

Standard DC1ds v3

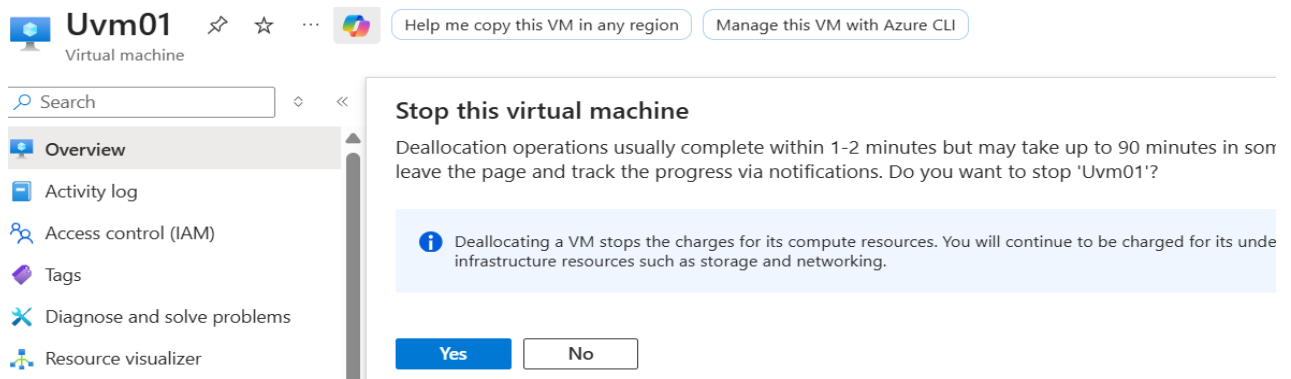
vCPUs

1

RAM

8 GiB

✓ **First Stop Machine and Click on Size Next select size and save.**



✓ **Next Start the machine and refresh then you can see changes below.**

Operating system : Linux  
Size : Standard DC2ds v3 (2 vcpus, 16 GiB memory)  
Primary NIC public IP : [23.100.85.96](#)  
[1 associated public IPs](#)  
Virtual network/subnet : [vnet-centralus/snet-centralus-1](#)  
DNS name : [Not configured](#)  
Health state : -  
Time created : 14/02/2026, 10:31 UTC

| Size  |                   |
|-------|-------------------|
| Size  | Standard DC2ds v3 |
| vCPUs | 2                 |
| RAM   | 16 GiB            |

✓ **We can also perform without stopping but in that case, it will take long time to upgrade.**

✓ Next for **Traffic distribution** we use **Load Balancer**

✓ ON WHAT BASES THE MACHINE IS GOING TO ADDED UP & DECREASE

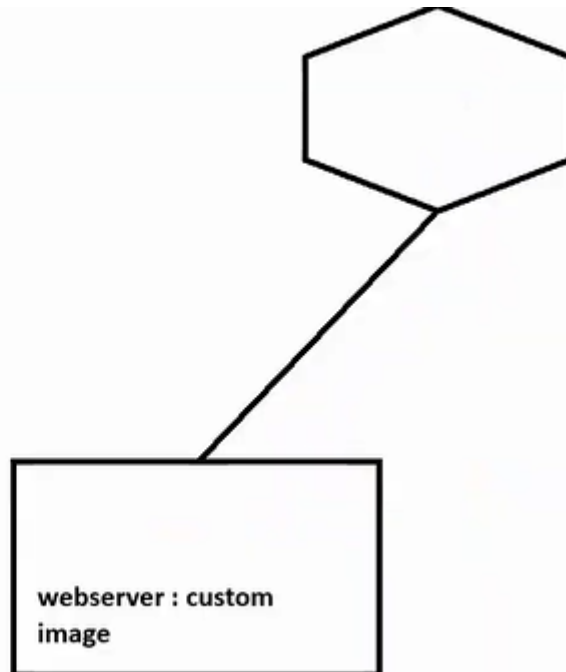
- Using **Scaling Condition**
- In Scaling condition → we use **CPU metric Condition**

Scaling condition

CPU metric condition

if CPU % is equal or above  
70% then 3 machine should  
add

if cpu % is equal or less 20%  
then 3 machines should  
delete



# Create a Virtual Machine Scale Set (VMSS) ...

## Scaling

Scaling mode ⓘ

- ☒ **Manually update the capacity:** Maintain a fixed amount of instances.
- ☐ **Autoscaling:** Scaling based on a CPU metric, on any schedule.
- ☐ **No scaling profile:** manual attach virtual machines after deployment

Instance count \* ⓘ

1 ✓

[Configure scaling options](#)

## Instance details

Image \* ⓘ

 Ubuntu Server 24.04 LTS - x64 Gen2 (free services eligible) ▼

[See all images](#) | [Configure VM generation](#)

VM architecture ⓘ

- ☐ Arm64
- ☒ x64

Size \* ⓘ

**Selected sizes**  
Standard D2ads v5  
[Select up to 5 sizes](#)

Enable Hibernation ⓘ

☐

 Hibernate does not currently support Trusted launch and Confidential virtual machines for Linux images. [Learn more](#) ↗

## Administrator account

Authentication type ⓘ

- ☒ Password
- ☐ SSH public key

# Create a Virtual Machine Scale Set (VMSS) ...

Subscription \*

Resource group \*

[Create new](#)

## Scale set details

Virtual machine scale set name \*

Region \*

[Deploy to an Azure Extended Zone](#)

Availability zone ⓘ

Autoscaling can help you respond to an outage by scaling out new instances in another zone.

## Orchestration

A scale set has a "scale set model" that defines the attributes of virtual machine instances (size, number of data disks, etc). As the number of instances in the scale set changes, new instances are added based on the scale set model.  
[Learn more about the scale set model](#)

Orchestration mode \* ⓘ

☒ **Flexible:** achieve high availability at scale with identical or multiple virtual machine types

☐ **Uniform:** optimized for large scale stateless workloads

Security type ⓘ

[Configure security features](#)

Home > Compute infrastructure | Virtual Machine Scale Set (VMSS)

## Create a Virtual Machine Scale Set (VMSS) ...

| NAME                     | CREATE PUBLI... | SUBNET                     | NETWORK SECU... | ACCELERATED N... |
|--------------------------|-----------------|----------------------------|-----------------|------------------|
| vnet-australiaeast-nic01 | Yes             | snet-australiaeast-1 (1... | Basic           | On               |

### Load balancing

You can place this virtual machine in the backend pool of an existing Azure load balancing solution. [Learn more](#)

Load balancing options ⓘ

- ☐ None
- ☒ **Azure load balancer**  
Supports all TCP/UDP network traffic, port-forwarding, and outbound flows.
- ☐ **Application gateway**  
Web traffic load balancer for HTTP/HTTPS with URL-based routing, SSL termination, session persistence, and web application firewall.

To allow traffic from your load balancing product, please update the appropriate port configuration on your network security group associated with your network interface.

Select a load balancer \* ⓘ

[Create a load balancer](#)

The value must not be empty.

[< Previous](#) [Next: Management >](#) [Review + create](#)

## Create a load balancer

Load balancer name \*

Type \* ⓘ

☒ **Public**  
Provides outbound connections for virtual machines inside your virtual network using public load balancers.

☐ **Internal**  
Used to load balance traffic inside a virtual network. A load balancer frontend can be accessed from an on-premises network in a hybrid scenario.

Protocol \* ⓘ

☒ TCP

☐ UDP

**Rules**

Rules

☒ Load balancer rule

☒ Inbound NAT rule

[Create](#) [Cancel](#)

# Create a Virtual Machine Scale Set (VMSS) ...

✓ Validation passed

BasicsSpotDisksNetworkingManagementHealthAdvancedTagsReview + create

Basics

Subscription

Azure subscription 1

Resource group

(new) UVmss-rg

Virtual machine scale set name

UVmss01

Region

Australia East

Orchestration mode

Flexible

Availability zone

1,2,3

Image

Ubuntu Server 24.04 LTS - Gen2

Size

Standard D2ads v5 (2 vcpus, 8 GiB memory)

Scaling mode

Manually update the capacity

Instance count

1

Security type

Trusted launch virtual machines

Enable secure boot

Yes

Enable vTPM

Yes

< Previous

Next >

Create

## Check VM & Load balancer

Home > CreateVmss-canonical.ubuntu-24\_04-lts-server-20260217160845 | Overview > UVmss01

UVmss01 | Instances

Virtual machine scale set

Search

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Instances

Resource visualizer

Manage view

Refresh

Export to CSV

Open query

Assign tags

Start

Restart

Stop

Hibernate

Group by

Filter for any field...

Subscription equals all

Status equals all

Add filter

|                          | Name ↑           | Computer name | Status  | Type | Provisioning state | Size              |
|--------------------------|------------------|---------------|---------|------|--------------------|-------------------|
| <input type="checkbox"/> | UVmss01_0fd58871 | ...           | Running | VM   | Succeeded          | Standard_D2ads_v5 |

UVmss01 | Network settings

Virtual machine scale set

Search

Diagnose and solve problems

Instances

Resource visualizer

Networking

Network settings

Load balancing

Application security groups

Network manager

Settings

Availability + scale

Scaling

Network security group basicNsgvnet-australiaeast-nic01

attached to networkInterface: vnet-australiaeast-nic01

Impacts 0 subnets, 2 network interfaces

Create port rule

Search rules

Source == all

Destination == all

Protocol == all

Action == all

Port == all

| Priority ↑              | Name                          | Port | Protocol | Source            | Destination    | Action |
|-------------------------|-------------------------------|------|----------|-------------------|----------------|--------|
| Inbound port rules (5)  |                               |      |          |                   |                |        |
| 100                     | allow                         | Any  | TCP      | Any               | Any            | Allow  |
| 300                     | Tcp                           | 80   | TCP      | Any               | Any            | Allow  |
| 65000                   | AllowVnetInBound              | Any  | Any      | VirtualNetwork    | VirtualNetwork | Allow  |
| 65001                   | AllowAzureLoadBalancerInBound | Any  | Any      | AzureLoadBalancer | Any            | Allow  |
| 65500                   | DenyAllInBound                | Any  | Any      | Any               | Any            | Deny   |
| Outbound port rules (3) |                               |      |          |                   |                |        |

Search

«

+ Create

Manage view

Refresh

Export to CSV

Open query

Assign tags

Add to service group

Group I

Filter for any field...

Subscription equals all

Resource Group equals all

Location equals all

Add filter

| <input type="checkbox"/> Name 1   | SKU      | Resource Group | Location       | Subscription         |
|-----------------------------------|----------|----------------|----------------|----------------------|
| <input type="checkbox"/> UdayLoad | Standard | UVmss-rg       | Australia East | Azure subscription 1 |

Home > UVmss01

## UVmss01 | Scaling

Virtual machine scale set

Search

«

Save

Discard

Refresh

Logs

Feedback

- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Instances
- Resource visualizer
- Networking
- Settings
- Availability + scale
- Scaling
- Availability
- Size
- Standby pool
- Security

Default \*

Default condition

Delete warning

The very last or default recurrence rule cannot be off autoscale.

Scale mode

Scale based on a metric

Scale to a specific number of instances

Rules

Scale out

When UVmss01 (Average)

Scale in

When UVmss01 (Average)

+ Add a rule

Instance limits

Minimum \*

2

Maximum \*

20

Schedule

This scale condition is executed when none of the following conditions are met.

+ Add a scale condition

## Scale rule

17:28

15

30

45

UTC+05:30

Percentage CPU (Maximum)

--

☐ Enable metric divide by instance count

Operator \*

Greater than or equal to

Metric threshold to trigger scale action \*

40

%

Duration (minutes) \*

1

Time grain (minutes)

1

Setting a duration less than 5 minutes may generate transient metric spikes that leads to unexpected scaling actions. For best results, the duration should be set at least to 5 minutes.

Time grain statistic \*

Average

Time aggregation \*

Maximum

Action

Operation \*

Increase count by

Cool down (minutes) \*

1

instance count \*

3

Update

Delete

Home > UVmss01

## UVmss01 | Scaling

Virtual machine scale set

Search

«

Save

Discard

Refresh

Logs

Feedback

- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Instances
- Resource visualizer
- Networking
- Settings
- Availability + scale
- Scaling
- Availability
- Size
- Standby pool
- Security

Default \*

Default condition

Delete warning

The very last or default recurrence rule cannot be off autoscale.

Scale mode

Scale based on a metric

Scale to a specific number of instances

Rules

Scale out

When UVmss01 (Maximum)

Scale in

When UVmss01 (Average)

+ Add a rule

Instance limits

Minimum \*

2

Maximum \*

20

Schedule

This scale condition is executed when none of the following conditions are met.

+ Add a scale condition

## Scale rule

Percentage CPU (Minimum)

15.23 %

☐ Enable metric divide by instance count

Operator \*

Less than or equal to

Metric threshold to trigger scale action \*

20

%

Duration (minutes) \*

1

Time grain (minutes)

1

Setting a duration less than 5 minutes may generate transient metric spikes that leads to unexpected scaling actions. For best results, the duration should be set at least to 5 minutes.

Time grain statistic \*

Average

Time aggregation \*

Minimum

Action

Operation \*

Decrease count by

Cool down (minutes) \*

1

instance count \*

3

Update

Delete

Add or remove favorites by pressing Ctrl+Shift+F



```

azureuser@uvmss01bu6EXQY1:~$ sudo su
root@uvmss01bu6EXQY1:/home/azureuser# apt update
Hit:1 http://azure.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://azure.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://azure.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 http://azure.archive.ubuntu.com/ubuntu noble-security InRelease [126 kB]

```

```

root@uvmss01bu6EXQY1:/home/azureuser# apt install stress -y

```

- Duplicate it and give **htop**

| Main |           | I/O |    |       |       |       |   |      |      |         |                                   |  |  |
|------|-----------|-----|----|-------|-------|-------|---|------|------|---------|-----------------------------------|--|--|
| PID  | USER      | PRI | NI | VRT   | RES   | SHR   | S | CPU% | MEM% | TIME+   | Command                           |  |  |
| 1954 | azureuser | 20  | 0  | 15260 | 8000  | 5796  | S | 0.0  | 0.1  | 0:00.88 | sshd: azureuser@pts/0             |  |  |
| 3080 | azureuser | 20  | 0  | 8744  | 4592  | 3544  | R | 0.7  | 0.1  | 0:05.76 | htop                              |  |  |
| 1    | root      | 20  | 0  | 22964 | 14184 | 9532  | S | 0.0  | 0.2  | 0:03.92 | /sbin/init                        |  |  |
| 138  | root      | 19  | -1 | 50436 | 11536 | 10340 | S | 0.0  | 0.1  | 0:00.51 | /usr/lib/systemd/systemd-journald |  |  |
| 199  | root      | RT  | 0  | 282M  | 27324 | 8760  | S | 0.0  | 0.3  | 0:00.06 | /sbin/multipathd -d -s            |  |  |
| 207  | root      | 20  | 0  | 282M  | 27324 | 8760  | S | 0.0  | 0.3  | 0:00.00 | /sbin/multipathd -d -s            |  |  |
| 209  | root      | RT  | 0  | 282M  | 27324 | 8760  | S | 0.0  | 0.3  | 0:00.00 | /sbin/multipathd -d -s            |  |  |
| 210  | root      | RT  | 0  | 282M  | 27324 | 8760  | S | 0.0  | 0.3  | 0:00.00 | /sbin/multipathd -d -s            |  |  |

```

root@uvmss01bu6EXQY1:/home/azureuser# stress
'stress' imposes certain types of compute stress on your system

```

```

Usage: stress [OPTION [ARG]] ...
  -?, --help            show this help statement
  --version            show version statement
  -v, --verbose         be verbose
  -q, --quiet          be quiet
  -n, --dry-run        show what would have been done
  -t, --timeout N      timeout after N seconds
  --backoff N          wait factor of N microseconds before work starts
  -c, --cpu N          spawn N workers spinning on sqrt()
  -i, --io N           spawn N workers spinning on sync()
  -m, --vm N           spawn N workers spinning on malloc()/free()
  --vm-bytes B         malloc B bytes per vm worker (default is 256MB)
  --vm-stride B        touch a byte every B bytes (default is 4096)
  --vm-hang N          sleep N secs before free (default none, 0 is inf)
  --vm-keep            redirty memory instead of freeing and reallocating
  -d, --hdd N          spawn N workers spinning on write()/unlink()
  --hdd-bytes B        write B bytes per hdd worker (default is 1GB)

```

```

Example: stress --cpu 8 --io 4 --vm 2 --vm-bytes 128M --timeout 10s

```

```

Example: stress --cpu 8 --io 4 --vm 2 --vm-bytes 128M --timeout 10s

```

Note: Numbers may be suffixed with s,m,h,d,y (time) or B,K,M,G (size).

```

root@uvmss01bu6EXQY1:/home/azureuser# ^C
root@uvmss01bu6EXQY1:/home/azureuser# stress --cpu 8 --io 4 --vm 2 --vm-bytes 128M --timeout 20m
stress: info: [3382] dispatching hogs: 8 cpu, 4 io, 2 vm, 0 hdd

```

- We can see traffic increasing 100% in duplicate
- Due to high traffic, machines getting created
- To stop traffic, click Ctrl+Z
- When we stop created machines will be deleted.